

Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number: Batch 13

Start date: 02-Feb-2026

Names: Sushant Pawar.

Internship Duration: 8 Weeks

1. Project Title

BankBot AI – Intelligent Chatbot for Banking FAQs

2. Project Objective

The main objective of this project is to develop an **AI-powered chatbot that can assist users with banking-related queries**. The chatbot acts as a virtual banking assistant capable of providing instant responses to frequently asked banking questions.

The system aims to improve digital banking support by reducing dependency on manual customer service while providing quick and accurate information to users.

This project also focuses on implementing **Artificial Intelligence, Natural Language Processing, and domain-restricted chatbot responses** to ensure that the chatbot provides relevant banking information only.

3. Project description in detail

The **BankBot AI Chatbot** is designed to provide automated responses to common banking queries using a conversational interface.

The system allows users to enter questions related to banking services such as ATM usage, account information, interest rates, and online banking. The chatbot processes the user input and generates appropriate responses.

The project was implemented using **Python and Streamlit**, which provides an interactive web interface for users to communicate with the chatbot.

The system uses a **hybrid approach** that combines a predefined FAQ knowledge base and an Artificial Intelligence model.

The chatbot first attempts to match the user's question with a stored FAQ database. If a match is found, the corresponding answer is returned instantly. If no suitable match is available, the system sends the query to an **AI model running locally using Ollama** to generate an intelligent response.

This project demonstrates how AI-powered virtual assistants can improve digital banking support systems by providing fast, automated, and reliable customer assistance.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Internship orientation and project planning	Understood project requirements and defined project scope
Week 2	Research on chatbot technologies and AI models	Explored AI chatbots, Ollama, and banking FAQ systems
Week 3	Design chatbot interface	Developed Streamlit-based chatbot UI
Week 4	Implement FAQ knowledge base	Created FAQ dataset and query matching logic
Week 5	Integrate AI model with chatbot	Integrated Ollama AI model for response generation
Week 6	Implement domain restriction	Added banking query filtering and guardrails
Week 7	Testing and debugging	Tested chatbot with various banking queries
Week 8	Final documentation and presentation	Prepared final report and project presentation

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Understanding project objectives and planning implementation	Week 1
Prototype/First Draft	Initial chatbot interface and FAQ system developed	Week 4
Mid-Term Review	AI model integration and testing completed	Week 6
Final Submission	Completed chatbot system and documentation	Week 8
Presentation	Demonstrated project functionality and results	Week 8

5b. Project execution details

Initially, research was conducted on chatbot technologies, AI models, and banking customer support systems. Based on the research, the project architecture was designed.

In the first phase, a chatbot user interface was developed using **Streamlit**, allowing users to interact with the system through a conversational interface.

In the second phase, a **FAQ knowledge base** was created to store common banking questions and answers. Query matching logic was implemented using keyword detection and similarity matching techniques.

Next, the chatbot was integrated with an **AI model using Ollama** to generate responses when the FAQ system could not provide an answer.

In the final stage, **domain restriction mechanisms** were implemented to ensure that the chatbot responds only to banking-related queries. The system was tested with multiple queries to verify functionality and accuracy.

6. Snapshots / Screenshots

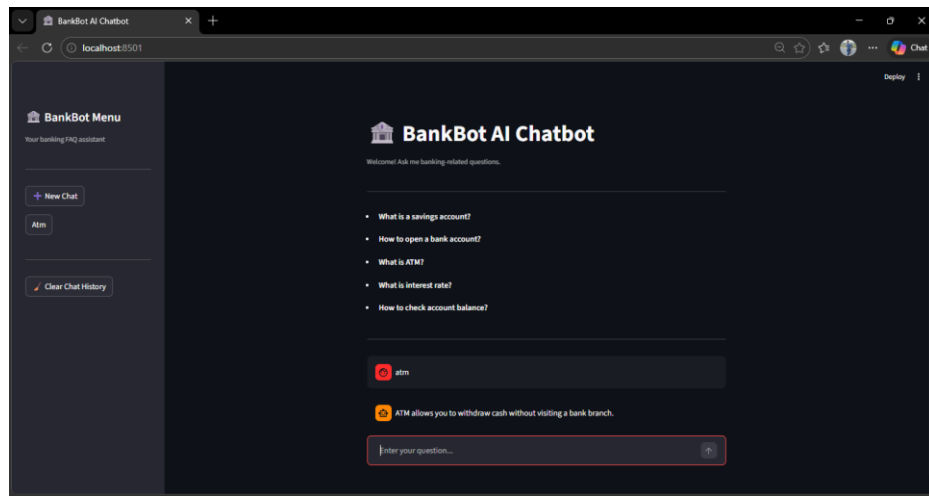


Fig. Chatbot user interface

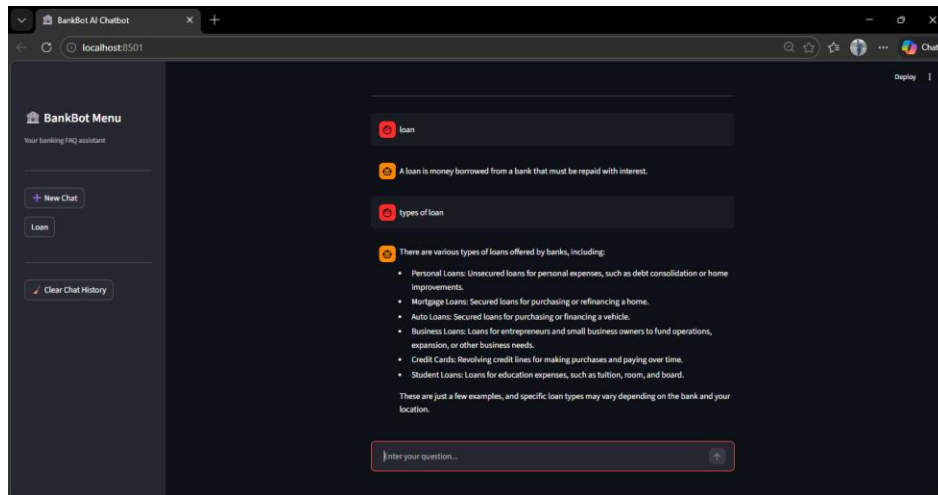


Fig. Chat interaction example

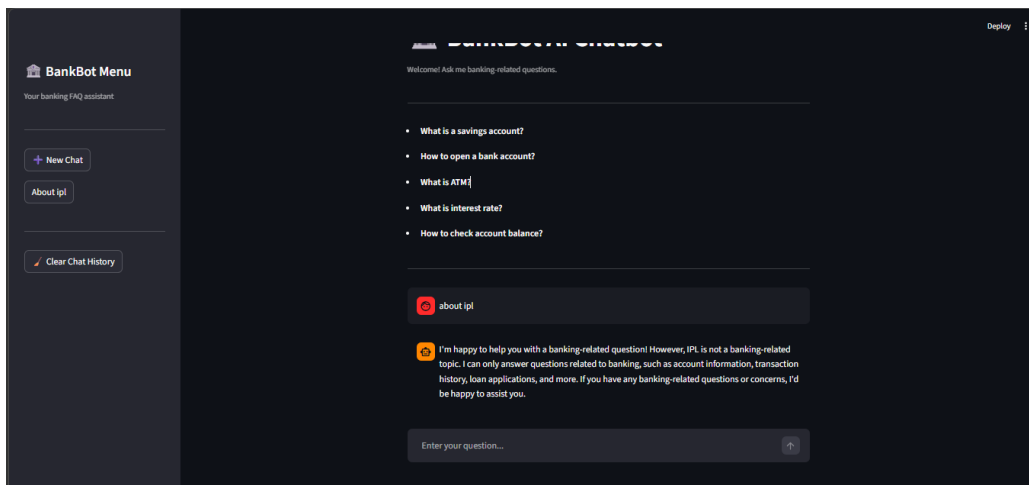


Fig. Chat interaction with domain restriction mechanisms

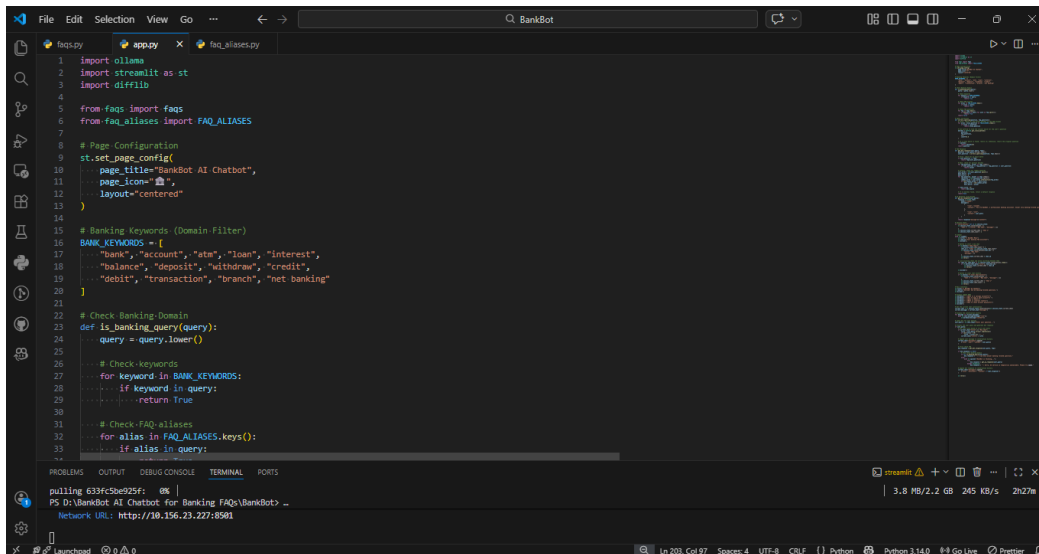


Fig. Code snippets of chatbot logic

7. Challenges Faced

During the development of the project, several challenges were encountered.

One challenge was ensuring that the chatbot correctly understands different variations of user queries. This was addressed by implementing **alias mapping and similarity matching techniques**.

Another challenge was controlling AI-generated responses to ensure that the chatbot only answers banking-related questions. This issue was resolved by implementing **domain restriction mechanisms and keyword filtering**.

Performance optimization was also considered. Initially, the Llama3 model was used, but due to slower response time it was replaced with the **Phi-3 model**, which provided faster responses.

These challenges helped improve the overall functionality and reliability of the chatbot system.

8. Learnings & Skills Acquired

Through this internship project, several technical and practical skills were gained.

Technical Skills:

- Python programming
- Streamlit web application development
- AI chatbot development
- Integration of AI models using Ollama
- Natural language query processing
- Domain restriction techniques

Soft Skills:

- Problem-solving
- Research and experimentation
- Project planning and execution
- Technical documentation

The internship provided valuable experience in building **AI-based applications for real-world use cases**.

9. Testimonials from team

This internship provided an excellent opportunity to work on a practical AI-based project.

Developing the BankBot AI Chatbot helped improve both technical and problem-solving skills. The project demonstrated how artificial intelligence can be applied to automate customer support in banking services.

The experience gained through this internship will be valuable for future projects and professional development in the field of artificial intelligence and software development.

10. Conclusion

The BankBot AI Chatbot project successfully demonstrates the implementation of an AI-powered virtual assistant for banking queries.

The system combines a **FAQ knowledge base, AI model integration, and domain restriction mechanisms** to provide fast and reliable responses to users.

This project highlights how AI chatbots can improve digital banking services by reducing customer support workload and providing instant assistance.

The internship provided practical experience in AI development and strengthened understanding of modern software technologies.

11. Acknowledgements

I would like to express my sincere gratitude to **Infosys Springboard** for providing this valuable internship opportunity.

I would also like to thank the mentors and program coordinators for their guidance, support, and learning resources throughout the internship.

Finally, I would like to acknowledge everyone who contributed to the successful completion of this internship project.